

The numerical range is a 2-spectral set

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Abstract

For an $n \times n$ complex matrix A , let $W(A) = \{x^*Ax : x \in \mathbb{C}^n, \|x\|_2 = 1\}$ be its numerical range. Here x^* denotes the conjugate transpose, $\|\cdot\|_2$ the Euclidean norm, and $\|\cdot\|$ the induced operator norm. For every function f holomorphic on a neighborhood of $W(A)$, we prove that

$$\|f(A)\| \leq 2 \max_{z \in W(A)} |f(z)|$$

and consequently that $W(A)$ is a 2-spectral set for A . This proves Crouzeix's conjecture, and the constant is optimal. The central ingredient is a positive-real completion theorem relative to an auxiliary eigenbasis: an adjoint-algebra constraint on the resolvent defect of a normalized matrix-valued Carathéodory function forces the underlying matrix to have norm at most 2. The classical positive double-layer calculus supplies such completions for $f(B)$ whenever the auxiliary matrix B has simple spectrum; the eigenvalues of $f(B)$ may repeat. Sampling the associated Herglotz kernel at half the conjugate diagonal entries and at the origin cancels the completion term's contribution and leaves a comparison of two weighted Gramians; their first nonconstant terms yield the sharp estimate. Simple-spectrum approximation and canonical convex outer domains with supported oriented radial boundaries yield the general theorem.

Keywords: Crouzeix conjecture; numerical range; spectral set; matrix-valued Herglotz kernel; double-layer potential

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1. Introduction

Let $\mathbb{C}$ denote the complex field and, for $n \geq 1$, let $\mathbb{M}_n(\mathbb{C})$ denote the algebra of $n \times n$ complex matrices. For $A \in \mathbb{M}_n(\mathbb{C})$, its numerical range is

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n, \|x\|_2 = 1\}. \quad (1)$$

Here x^* denotes the conjugate transpose, $\sigma(A)$ denotes the spectrum, $\|x\|_2$ is the Euclidean norm, $\|X\|$ is the induced operator norm, and scalar moduli are denoted by $|\cdot|$. The set $W(A)$ is compact, the Toeplitz–Hausdorff theorem makes it convex, and $\sigma(A) \subset W(A)$. Functions holomorphic near $\sigma(A)$ are evaluated using the holomorphic functional calculus.

A compact set $K \subset \mathbb{C}$ containing $\sigma(A)$ is called a C -spectral set for A if

$$\|r(A)\| \leq C \max_{z \in K} |r(z)| \quad (2)$$

for every rational function r pole-free on K . Our main result settles Crouzeix's conjecture by identifying the sharp universal constant for $K = W(A)$.

Theorem 1.1 (Main theorem). *Let $n \geq 1$, let $A \in \mathbb{M}_n(\mathbb{C})$, and let f be holomorphic on a neighborhood of $W(A)$. Then*

$$\|f(A)\| \leq 2 \max_{z \in W(A)} |f(z)|. \quad (3)$$

Consequently, $W(A)$ is a 2-spectral set for A .

The factor 2 is optimal: for

$$J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad p(z) = z, \quad (4)$$

one has $\max_{z \in W(J)} |z| = 1/2$ and $\|J\| = 1$.

For later use, write

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}.$$

We use I for the identity matrix of the size dictated by context, X^* for the adjoint, put $\operatorname{Re} X = (X + X^*)/2$, and let $\operatorname{alg}(T)$ denote the unital algebra generated by T . For Hermitian matrices X and Y , we write $X \preceq Y$ (equivalently, $Y \succeq X$) when $Y - X$ is positive semidefinite.

Crouzeix proved the estimate with constant 2 for 2×2 matrices and formulated the general conjecture in [1]. His subsequent dimension-free theorem gave the constant 11.08 [2]. After further structural and numerical work [3], Crouzeix and Palencia proved that $W(A)$ is a $(1 + \sqrt{2})$ -spectral set [4]. Their double-layer and Cauchy-transform argument was clarified in [5] and developed in several directions in [6,7].

The intervening literature established the conjectured constant for perturbed Jordan blocks and, more generally, scalar translates of cyclic weighted shifts, as well as for tridiagonal 3×3 matrices with elliptic numerical range centered at an eigenvalue [8–10]. Numerical and extremal investigations appear in [11,12], and a broad account of the problem and its variants is given in [13]. Subsequent work has treated Blaschke-product level sets, cyclicity reductions, compressions of shifts, further nilpotent classes, and quantitative bounds for KMS matrices [14–17]. An alternative proof for a class of weighted shifts appears in [18]. Across these cited reductions, the best dimension-free universal bound remained $1 + \sqrt{2}$.

1.1. The sharpness barrier in the symmetrized calculus

The positive boundary measure used below originates in the convex-domain integral calculus of Delyon–Delyon [19]; see also [4,20]. If B has numerical range in a convex domain Ω with continuously differentiable boundary and f is holomorphic on a neighborhood of $\overline{\Omega}$ with $|f| \leq 1$ on $\overline{\Omega}$, this calculus produces a unital positive map Φ and a holomorphic Cauchy companion g_f such that

$$2\Phi(f) = f(B) + g_f(B)^*. \quad (5)$$

Thus positivity directly controls a symmetrized functional calculus, while the target estimate concerns $f(B)$. The Crouzeix–Palencia argument couples the two terms in (5); the example in [5] shows that its abstract norm estimate is sharp at $1 + \sqrt{2}$. The corresponding general uniform-algebra theorem has the same sharp constant; its additional unital case was proposed as a possible route to 2 in [21]. Schwenninger and de Vries revisited the double-layer method [22]. Malman et al. proved that a configuration-constant refinement gives a strict improvement for every fixed numerical-range shape; they also constructed thin quadrilaterals for which the resulting bounds approach $1 + \sqrt{2}$ [23]. Separately, continuity and compactness give a constant $C_n < 1 + \sqrt{2}$ in every fixed matrix dimension n [24].

The key additional information is algebraic. For the approximants constructed in Section 5, the companion in (5) belongs to the algebra generated by B^* , where the simple-spectrum matrix B supplies

an eigenbasis that also diagonalizes $f(B)$. Treating $f(B)$ and the companion jointly preserves this coupling.

1.2. Auxiliary-basis de-symmetrization

Our argument uses the whole Cayley family associated with (5). It has four steps.

1. Applying Φ to

$$\zeta \mapsto \frac{1 + wf(\zeta)}{1 - wf(\zeta)}, \quad w \in \mathbb{D},$$

produces a matrix-valued Carathéodory function H , meaning that $H(0) = I$ and $\operatorname{Re} H(w) \succeq 0$.

2. For a simple-spectrum auxiliary matrix B , put $T = f(B)$. The double-layer identity gives the positive-real completion

$$H(w) - (I - wT)^{-1} \in \operatorname{alg}(B^*), \quad w \in \mathbb{D}. \quad (6)$$

3. The eigenbasis of B diagonalizes $T = f(B)$ even when some values of f on the spectrum of B coincide, and it turns the unknown term in (6) into a diagonal analytic correction. We sample the Herglotz kernel of H at half the conjugate diagonal entries of T and add one vector sample at the origin. Repeated sampling points are allowed, and the origin sample cancels the diagonal correction's contribution exactly.

4. The surviving positive-matrix inequality compares two weighted Gramians. Its difference is positive, and an eigenvector argument followed by the first nonconstant Gramian term yields $\|T\| \leq 2$.

The proof uses two limits: simple-spectrum matrices tend to A , followed by outer domains converging to $W(A)$. Together they yield the general case.

2. Positive-real kernels

Let $\mathcal{K}$ be an $\mathbb{M}_n(\mathbb{C})$ -valued kernel on $\mathbb{D} \times \mathbb{D}$. We call $\mathcal{K}$ positive if, for every finite choice $w_0, \dots, w_m \in \mathbb{D}$ and $\xi_0, \dots, \xi_m \in \mathbb{C}^n$,

$$\sum_{i,j=0}^m \xi_i^* \mathcal{K}(w_i, w_j) \xi_j \geq 0. \quad (7)$$

We shall use the following standard matrix-valued Herglotz kernel fact. A direct proof is included to fix the normalization.

Lemma 2.1 (Matrix Herglotz kernel). *Let $F: \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ be analytic with $\operatorname{Re} F(w) \succeq 0$ for every $w \in \mathbb{D}$. Then*

$$\mathcal{L}_F(w, z) = \frac{F(w) + F(z)^*}{1 - w\bar{z}} \quad (8)$$

is a positive kernel.

Proof. Fix $w_0, \dots, w_m \in \mathbb{D}$ and $\xi_0, \dots, \xi_m \in \mathbb{C}^n$, and put

$$\eta(\zeta) = \sum_{j=0}^m \frac{\xi_j}{1 - \bar{w}_j \zeta}, \quad |\zeta| = 1.$$

For $0 \leq r < 1$, the scalar Cauchy formula, applied entrywise, gives

$$\frac{1}{2\pi} \int_0^{2\pi} \eta(e^{it})^* 2 \operatorname{Re} F(re^{it}) \eta(e^{it}) dt = \sum_{i,j=0}^m \xi_i^* \frac{F(rw_i) + F(rw_j)^*}{1 - w_i \bar{w}_j} \xi_j. \quad (9)$$

The integrand on the left is nonnegative, so the right side is nonnegative. Letting $r \uparrow 1$ and using the continuity of F at the finitely many points w_i gives (7) for $\mathcal{K} = \mathcal{L}_F$. $\square$

3. The positive-real completion principle

The following completion theorem is the core of the argument: it converts a positive-real completion constrained by the adjoint algebra of an auxiliary simple-spectrum matrix into the sharp norm bound.

Theorem 3.1 (Positive-real completion). *Let*

$$B = S \operatorname{diag}(\beta_1, \dots, \beta_n) S^{-1}, \quad T = S \Lambda S^{-1}, \quad \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n), \quad (10)$$

where S is invertible, the numbers $\beta_1, \dots, \beta_n$ are distinct, and $|\lambda_i| \leq 1$ for every i . The numbers λ_i may repeat. Let $H: \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ be analytic and satisfy

$$H(0) = I, \quad \operatorname{Re} H(w) \succeq 0 \quad (w \in \mathbb{D}), \quad (11)$$

and

$$H(w) - (I - wT)^{-1} \in \operatorname{alg}(B^*) \quad (w \in \mathbb{D}). \quad (12)$$

Then $\|T\| \leq 2$.

Proof. Use the simultaneous diagonalizations in (10) and put $G = S^* S \succ 0$. Since the β_i are distinct, polynomial interpolation shows that

$$\operatorname{alg}(B^*) = \left\{ (S^{-1})^* D S^* : D \text{ is diagonal} \right\}. \quad (13)$$

By (13), the matrix

$$\Theta(w) = S^* (H(w) - (I - wT)^{-1}) (S^*)^{-1}$$

is diagonal for every w , and

$$H(w) - (I - wT)^{-1} = (S^{-1})^* \Theta(w) S^*.$$

Also $\Theta(0) = 0$ because $H(0) = I$. Write $\Theta(w) = \operatorname{diag}(\theta_1(w), \dots, \theta_n(w))$. On setting $\tilde{H}(w) = S^* H(w) S$, we obtain

$$\tilde{H}(w) = G(I - w\Lambda)^{-1} + \Theta(w)G. \quad (14)$$

Since $\operatorname{Re} \tilde{H}(w) = S^* (\operatorname{Re} H(w)) S$, the real part of $\tilde{H}(w)$ is positive semidefinite. Consequently, Lemma 2.1 applies to $\tilde{H}$.

Define Hermitian matrices $P, Q, Y \in \mathbb{M}_n(\mathbb{C})$ by

$$P_{ij} = \frac{G_{ij}}{1 - \overline{\lambda_i} \lambda_j / 4}, \quad Q_{ij} = \frac{G_{ij}}{1 - \overline{\lambda_i} \lambda_j / 2}, \quad Y = Q - P. \quad (15)$$

The denominators are nonzero because $|\lambda_i| \leq 1$. Since $G = G^*$, conjugate symmetry in the displayed formulas also gives $P = P^*$ and $Q = Q^*$. Let $e_1, \dots, e_n$ be the standard basis of $\mathbb{C}^n$, and fix $u = (u_1, \dots, u_n)^T \in \mathbb{C}^n$. In the positive-kernel inequality use the n points and vectors

$$w_i = \frac{\overline{\lambda_i}}{2}, \quad \xi_i = u_i e_i \quad (1 \leq i \leq n), \quad (16)$$

and add the point $w_0 = 0$ with vector

$$\xi_0 = v = -G^{-1} P u. \quad (17)$$

The points w_i may coincide. Kernel positivity applies to arbitrary finite indexed lists of points and vectors, including $w_i = w_0$ when $\lambda_i = 0$. We first verify that this sample cancels the contribution of Θ to the sampled kernel quadratic form. The contribution of the second term in (14) to the kernel quadratic form is

$$2 \operatorname{Re} \sum_{i=1}^n \overline{u_i} \theta_i(w_i) \left((Gv)_i + \sum_{j=1}^n \frac{G_{ij} u_j}{1 - w_i \overline{w_j}} \right). \quad (18)$$

Because

$$1 - w_i \overline{w_j} = 1 - \overline{\lambda_i} \lambda_j / 4,$$

the parenthesis in (18) is $(Gv + Pu)_i$, which vanishes by (17).

It remains to calculate the contribution of $G(I - w\Lambda)^{-1}$. For sample indices $r, s \in \{1, \dots, n\}$, compression of the kernel block between w_r and w_s against e_r and e_s gives

$$\frac{2G_{rs}}{(1 - \overline{\lambda_r} \lambda_s / 2)(1 - \overline{\lambda_r} \lambda_s / 4)} = (4Q - 2P)_{rs}. \quad (19)$$

After compression by the coordinate vectors e_i , the two effective sample–origin blocks are $G + Q$, and the origin block is $2G$. After the cancellation of Θ 's contribution, kernel positivity gives, for the prescribed vector $v = -G^{-1}Pu$,

$$\begin{aligned} 0 &\leq \begin{pmatrix} u \\ v \end{pmatrix}^* \begin{pmatrix} 4Q - 2P & G + Q \\ G + Q & 2G \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} \\ &= u^* ((4Q - 2P) - (G + Q)G^{-1}P - PG^{-1}(G + Q) + 2PG^{-1}P)u \\ &= u^* (4Y - YG^{-1}P - PG^{-1}Y)u, \end{aligned} \quad (20)$$

Here the block quadratic form is used only on the graph $v = -G^{-1}Pu$; it is not asserted to be positive for independent u and v . The last equality uses $Q = P + Y$. Since u was arbitrary,

$$4Y - YG^{-1}P - PG^{-1}Y \succeq 0. \quad (21)$$

Balance the diagonalization by defining

$$\begin{aligned} \tilde{T} &= G^{1/2} \Lambda G^{-1/2}, & \hat{P} &= G^{-1/2} P G^{-1/2}, \\ \hat{Q} &= G^{-1/2} Q G^{-1/2}, & \hat{Y} &= \hat{Q} - \hat{P}. \end{aligned} \quad (22)$$

Expanding the scalar kernels in (15) into geometric series gives

$$\hat{P} = \sum_{k=0}^{\infty} 4^{-k} \tilde{T}^{*k} \tilde{T}^k, \quad \hat{Q} = \sum_{k=0}^{\infty} 2^{-k} \tilde{T}^{*k} \tilde{T}^k. \quad (23)$$

Indeed, $\tilde{T}^k = G^{1/2} \Lambda^k G^{-1/2}$, so the sequence $(\tilde{T}^k)$ is bounded even when some $|\lambda_j| = 1$; the geometric weights give norm convergence. It follows that

$$\hat{Y} = \sum_{k=1}^{\infty} (2^{-k} - 4^{-k}) \tilde{T}^{*k} \tilde{T}^k \succeq 0. \quad (24)$$

Taking the congruence of (21) by $G^{-1/2}$ yields

$$4\hat{Y} - \hat{Y}\hat{P} - \hat{P}\hat{Y} \succeq 0. \quad (25)$$

Let e be an eigenvector of the Hermitian matrix $\hat{P}$, say $\hat{P}e = \alpha e$. If $\alpha > 2$, then (25) gives

$$0 \leq 2(2 - \alpha)e^* \hat{Y} e. \quad (26)$$

Since $\hat{Y} \succeq 0$ and $\alpha > 2$, (26) forces $e^* \hat{Y} e = 0$. The $k = 1$ term in (24) is $\tilde{T}^* \tilde{T} / 4$, so $\tilde{T} e = 0$. The first series in (23) gives $\hat{P} e = e$, contradicting $\hat{P} e = \alpha e$ with $\alpha > 2$. The first series in (23) also begins with I , and hence

$$I \preceq \hat{P} \preceq 2I. \quad (27)$$

The $k = 1$ term of the same series now gives

$$\frac{1}{4} \tilde{T}^* \tilde{T} \preceq \hat{P} - I \preceq I.$$

Thus $\|\tilde{T}\| \leq 2$. The polar decomposition $S = UG^{1/2}$ gives

$$T = U \tilde{T} U^*,$$

where U is unitary. Therefore $\|T\| \leq 2$. $\square$

Remark 3.2. The factor $1/2$ in the sampling points (16) simultaneously produces the weights $1/2$ and $1/4$ in (23). Their positive difference bounds $\hat{P}$, and the first nonconstant term of $\hat{P}$ then gives the sharp constant. The origin sample cancels the correction's contribution by an exact algebraic identity. Distinct auxiliary eigenvalues identify $\text{alg}(B^*)$ with the diagonal algebra in the chosen basis, while the diagonal entries of T may repeat.

4. The double-layer completion

We now construct from the numerical range the function required by Theorem 3.1. The measure and companion transform used here are classical [2,4,19]. The full Cayley family yields the algebraic constraint (12) relative to the auxiliary matrix.

Only the canonical parallel domains used in Section 5 are needed. We say that a bounded convex domain Ω has a *supported oriented radial boundary* if, for some $c \in \Omega$, its boundary is traced once by

$$\gamma(t) = c + \rho(t)e^{it}, \quad 0 \leq t \leq 2\pi,$$

where ρ is positive, continuously differentiable, and 2π -periodic, and there are continuous 2π -periodic functions v and s such that $|v(t)| = 1$, $s(t) \geq 0$,

$$\gamma'(t) = iv(t)s(t), \quad \text{Re}(\overline{v(t)}(\gamma(t) - z)) \geq 0 \quad (z \in \Omega). \quad (28)$$

Thus v is the outward unit normal, $s(t) dt$ is arclength, and the orientation is positive. Lemma 5.2 below supplies exactly this boundary for every canonical parallel domain.

Lemma 4.1 (Radial double-layer positive map). *Let $B \in \mathbb{M}_n(\mathbb{C})$, let $\Omega \subset \mathbb{C}$ be a bounded convex domain with a supported oriented radial boundary γ , and suppose $W(B) \subset \Omega$. There is a bounded complex-linear, unital, positive map on the compact parameter interval,*

$$\Phi_\gamma: C([0, 2\pi]) \longrightarrow \mathbb{M}_n(\mathbb{C}),$$

given by the double-layer density below.

Proof. Use the data in (28), and put

$$Z_t = \gamma(t)I - B, \quad \mathcal{R}_t = Z_t^{-1}.$$

This resolvent exists because $\sigma(B) \subset W(B) \subset \Omega$. Define the matrix-valued boundary density

$$D_B(t) = \frac{s(t)}{2\pi} \left(v(t)\mathcal{R}_t + \overline{v(t)}\mathcal{R}_t^* \right). \quad (29)$$

It is positive semidefinite. The support inequality in (28) gives

$$v(t)Z_t^* + \overline{v(t)}Z_t = 2\operatorname{Re}(\overline{v(t)}Z_t) \succeq 0. \quad (30)$$

Indeed, for every unit vector $x \in \mathbb{C}^n$, the quadratic form on the left is

$$2\operatorname{Re}(\overline{v(t)}(\gamma(t) - x^*Bx)) \geq 0,$$

because $x^*Bx \in W(B) \subset \Omega$. Congruence by $\mathcal{R}_t$ turns the left side of (30) into the matrix in parentheses in (29).

Since $d\gamma = iv(t)s(t)dt$, the matrix Cauchy formula gives

$$\frac{1}{2\pi} \int_0^{2\pi} v(t)\mathcal{R}_t s(t) dt = I. \quad (31)$$

Taking adjoints gives the same identity for the second layer. Hence

$$\int_0^{2\pi} D_B(t) dt = 2I. \quad (32)$$

For $\varphi \in C([0, 2\pi])$, define

$$\Phi_\gamma(\varphi) = \frac{1}{2} \int_0^{2\pi} \varphi(t) D_B(t) dt. \quad (33)$$

The positivity of D_B and (32) make Φ_γ positive and unital, hence $*$ -preserving. It is bounded because

$$\|\Phi_\gamma(\varphi)\| \leq \frac{1}{2} \|\varphi\|_\infty \int_0^{2\pi} \|D_B(t)\| dt.$$

□

Proposition 4.2 (Radial Cayley completion). *Let Ω be a bounded convex domain with a supported oriented radial boundary, let $B \in \mathbb{M}_n(\mathbb{C})$ have simple spectrum, and suppose $W(B) \subset \Omega$. Let V be an open neighborhood of $\overline{\Omega}$, and let f be holomorphic on V and satisfy*

$$\max_{\zeta \in \overline{\Omega}} |f(\zeta)| \leq 1. \quad (34)$$

Put $T = f(B)$. Then there is an analytic $H: \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ satisfying

$$H(0) = I, \quad \operatorname{Re} H(w) \succeq 0 \quad (w \in \mathbb{D}), \quad (35)$$

and

$$H(w) - (I - wT)^{-1} \in \operatorname{alg}(B^*) \quad (w \in \mathbb{D}). \quad (36)$$

Proof. Choose

$$B = S \operatorname{diag}(\beta_1, \dots, \beta_n) S^{-1}.$$

Then $T = S \operatorname{diag}(f(\beta_1), \dots, f(\beta_n)) S^{-1}$, and (34) gives $|f(\beta_i)| \leq 1$ for every i . In particular, $\sigma(T) \subset \overline{\mathbb{D}}$.

For the fixed boundary data, put $\mathcal{R}_t = (\gamma(t)I - B)^{-1}$, and let Φ_γ be the bounded map from Lemma 4.1. For $w \in \mathbb{D}$, define

$$c_w(t) = \frac{1 + wf(\gamma(t))}{1 - wf(\gamma(t))} \quad (37)$$

and put

$$H(w) = \Phi_\gamma(c_w). \quad (38)$$

The expansion

$$c_w(t) = 1 + 2 \sum_{m=1}^{\infty} w^m f(\gamma(t))^m \quad (39)$$

converges locally uniformly in $C([0, 2\pi])$. The boundedness of Φ_γ therefore makes H analytic, and $H(0) = I$. Furthermore,

$$\operatorname{Re} c_w(t) = \frac{1 - |wf(\gamma(t))|^2}{|1 - wf(\gamma(t))|^2} \geq 0.$$

The positivity and $*$ -preservation of Φ_γ give $\operatorname{Re} H(w) = \Phi_\gamma(\operatorname{Re} c_w) \succeq 0$.

For every $m \geq 0$, the scalar Cauchy formula applied in the chosen eigenbasis gives exactly

$$\frac{1}{2\pi} \int_0^{2\pi} f(\gamma(t))^m v(t) \mathcal{R}_t s(t) dt = T^m. \quad (40)$$

Indeed, this is the diagonal collection of the scalar Cauchy formulas for f^m at the points β_i ; the neighborhood V contains the contour and its interior. Transporting the uniformly convergent series (39) through this first-layer integral gives

$$I + 2 \sum_{m=1}^{\infty} w^m T^m = (I + wT)(I - wT)^{-1}.$$

The matrix series converges because the powers of T are bounded in the displayed eigenbasis.

Define the companion matrix

$$G(w) = \frac{1}{2\pi} \int_0^{2\pi} \overline{c_w(t)} v(t) \mathcal{R}_t s(t) dt.$$

Splitting the density (29) into its two layers now gives

$$2H(w) = (I + wT)(I - wT)^{-1} + G(w)^*.$$

Each $\mathcal{R}_t$ belongs to $\operatorname{alg}(B)$, and this finite-dimensional algebra is closed, so $G(w) \in \operatorname{alg}(B)$. Since $(I + wT)(I - wT)^{-1} = 2(I - wT)^{-1} - I$, we conclude that

$$H(w) - (I - wT)^{-1} = \frac{1}{2}(G(w)^* - I) \in \operatorname{alg}(B^*). \quad (41)$$

This proves (36). $\square$

Corollary 4.3 (Auxiliary sharp bound). *Under the hypotheses of Proposition 4.2, $\|f(B)\| \leq 2$.*

Proof. Write

$$B = S \operatorname{diag}(\beta_1, \dots, \beta_n) S^{-1}.$$

Then

$$T = f(B) = S \operatorname{diag}(f(\beta_1), \dots, f(\beta_n)) S^{-1}.$$

Since $\sigma(B) \subset \Omega$, the bound on f gives $|f(\beta_i)| \leq 1$ for every i . Apply Theorem 3.1 to these simultaneous diagonalizations and to the function H constructed in Proposition 4.2. The values $f(\beta_i)$ may repeat. $\square$

5. Passage to arbitrary matrices and proof of the main theorem

Successive approximations pass from the auxiliary setting to arbitrary matrices.

Lemma 5.1 (Simple spectrum and numerical range). *Matrices with n distinct eigenvalues are dense in $\mathbb{M}_n(\mathbb{C})$. Moreover, for $A, B \in \mathbb{M}_n(\mathbb{C})$,*

$$W(B) \subset W(A) + \{z \in \mathbb{C} : |z| \leq \|B - A\|\}. \quad (42)$$

Proof. The discriminant of the characteristic polynomial is a nonzero polynomial in the matrix entries, as is seen at $\text{diag}(1, \dots, n)$. The zero set of a nonzero complex polynomial has empty interior, so its nonvanishing set is dense in $\mathbb{M}_n(\mathbb{C})$, proving the first assertion. For a unit vector x ,

$$|x^*(B - A)x| \leq \|B - A\|,$$

which proves (42). $\square$

Lemma 5.2 (Parallel convex domains). *Let $K \subset \mathbb{C}$ be nonempty, compact, and convex. For $\delta > 0$, the set*

$$\Omega_\delta = \{z \in \mathbb{C} : \text{dist}(z, K) < \delta\}$$

is a bounded convex domain, $\overline{\Omega}_\delta = K + \delta\overline{\mathbb{D}}$, and Ω_δ has a supported oriented radial boundary.

Proof. The set Ω_δ is bounded and convex, and its closure is the Minkowski sum $K + \delta\overline{\mathbb{D}}$. Identify $\mathbb{C}$ with $\mathbb{R}^2$. The metric projection $\pi_K: \mathbb{C} \rightarrow K$ is unique and continuous. Put $d_K(z) = \text{dist}(z, K)$. For $z, h \in \mathbb{C}$, comparison first with $\pi_K(z)$ and then with $\pi_K(z + h)$ gives

$$\begin{aligned} 2\text{Re}(\overline{z + h - \pi_K(z + h)}h) - |h|^2 &\leq d_K(z + h)^2 - d_K(z)^2 \\ &\leq 2\text{Re}(\overline{z - \pi_K(z)}h) + |h|^2. \end{aligned}$$

The continuity of π_K now shows that d_K^2 is continuously differentiable, with real gradient $2(z - \pi_K(z))$. On the level set $d_K(z) = \delta$ this gradient has norm $2\delta > 0$. Since $\partial\Omega_\delta = \{d_K = \delta\}$, the implicit-function theorem shows that $\partial\Omega_\delta$ is continuously differentiable.

Choose $c \in K$. The ball $c + \delta\overline{\mathbb{D}}$ lies in Ω_δ , so every ray from c meets the boundary once and transversely. Its radial function ρ is therefore positive, continuously differentiable, and periodic. With

$$\nu(t) = \frac{\gamma(t) - \pi_K(\gamma(t))}{\delta} \quad (0 \leq t \leq 2\pi)$$

as the outward unit normal along γ and $s(t) = |\gamma'(t)|$, convexity gives the support inequality in (28); positive orientation gives $\gamma'(t) = i\nu(t)s(t)$. These data form the required supported oriented radial boundary. $\square$

Proof of Theorem 1.1. Put $K = W(A)$, and let U be an open neighborhood of K on which f is holomorphic. Choose $\varepsilon_0 > 0$ such that the closed ε_0 -neighborhood of K lies in U . For $0 < \varepsilon < \varepsilon_0$, let

$$\Omega_\varepsilon = \{z \in \mathbb{C} : \text{dist}(z, K) < \varepsilon\}.$$

Lemma 5.2 makes this a bounded convex domain with a supported oriented radial boundary. The open convex set $V = \Omega_{\varepsilon_0}$ satisfies $\overline{\Omega}_\varepsilon \subset V \subset U$.

By Lemma 5.1, choose simple-spectrum matrices $B_k \rightarrow A$. For fixed ε , all sufficiently large k satisfy $\|B_k - A\| < \varepsilon/2$, and then (42) gives

$$W(B_k) \subset K + (\varepsilon/2)\overline{\mathbb{D}} \subset \Omega_\varepsilon.$$

Set

$$m_\varepsilon = \max_{z \in \overline{\Omega}_\varepsilon} |f(z)|.$$

If $m_\varepsilon = 0$, then f vanishes on a neighborhood of $\sigma(A)$, so $f(A) = 0$ and there is nothing to prove. Otherwise, apply Corollary 4.3 to B_k and f/m_ε to obtain

$$\|f(B_k)\| \leq 2m_\varepsilon.$$

Continuity of the holomorphic functional calculus, obtained from a fixed contour inside U , gives $f(B_k) \rightarrow f(A)$. Hence

$$\|f(A)\| \leq 2m_\varepsilon. \quad (43)$$

Put $m_0 = \max_{z \in K} |f(z)|$. Uniform continuity of f on the closed ε_0 -neighborhood of K gives a modulus of continuity ω_f there. Since every point of $\overline{\Omega}_\varepsilon$ lies within ε of K ,

$$0 \leq m_\varepsilon - m_0 \leq \omega_f(\varepsilon).$$

Letting $\varepsilon \downarrow 0$ in (43) proves (3).

Finally, a rational function pole-free on the compact set $W(A)$ is holomorphic on a neighborhood of $W(A)$. Applying (3) to such functions proves (2) with $K = W(A)$ and $C = 2$. $\square$

6. Consequences and scope

The Hilbert-space extension follows by the standard finite-dimensional compression argument from [2]; compare the survey [13].

Corollary 6.1 (Hilbert-space form). *Let $\mathcal{H} \neq \{0\}$ be a complex Hilbert space, with inner product linear in the first variable, and let $\mathcal{B}(\mathcal{H})$ denote its algebra of bounded operators. For $A \in \mathcal{B}(\mathcal{H})$, put*

$$W(A) = \{\langle Ax, x \rangle : x \in \mathcal{H}, \|x\| = 1\}.$$

Then, for every $p \in \mathbb{C}[z]$,

$$\|p(A)\| \leq 2 \sup_{z \in W(A)} |p(z)|. \quad (44)$$

Consequently, $\overline{W(A)}$ is a 2-spectral set for A .

Proof. The assertion is immediate if $p = 0$. Otherwise, write $d = \deg p$. For a unit vector $x \in \mathcal{H}$, let

$$\mathcal{M}_x = \text{span}\{x, Ax, \dots, A^d x\}, \quad A_x = \Pi_x A|_{\mathcal{M}_x},$$

where Π_x is the orthogonal projection onto $\mathcal{M}_x$. Induction gives

$$A_x^k x = A^k x \quad (0 \leq k \leq d), \quad (45)$$

because $A^{k+1}x \in \mathcal{M}_x$ whenever $k < d$. Moreover, $W(A_x) \subset W(A)$. Theorem 1.1, applied on the finite-dimensional space $\mathcal{M}_x$, therefore gives

$$\|p(A)x\|_{\mathcal{H}} = \|p(A_x)x\|_{\mathcal{H}} \leq \|p(A_x)\| \leq 2 \sup_{z \in W(A)} |p(z)|.$$

Taking the supremum over unit vectors x proves (44).

Put $K = \overline{W(A)}$. The standard inclusion $\sigma(A) \subset K$ will also be used below. Let $r = a/b$ be a reduced rational function with no pole on K . Then b has no zero on K , so spectral mapping and $\sigma(A) \subset K$ make $b(A)$ invertible.

The complement of the compact convex set K is connected. Polynomial Runge approximation therefore gives polynomials p_j converging uniformly to r on K . By (44),

$$\|p_j(A) - p_k(A)\| \leq 2 \max_{z \in K} |p_j(z) - p_k(z)|,$$

so $p_j(A)$ converges in operator norm to some R , with

$$\|R\| \leq 2 \max_{z \in K} |r(z)|.$$

Since $p_j b \rightarrow a$ uniformly on K , the same polynomial estimate gives

$$p_j(A)b(A) = (p_j b)(A) \rightarrow a(A).$$

Thus $Rb(A) = a(A)$ and $R = a(A)b(A)^{-1} = r(A)$. This proves the rational estimate and hence the spectral-set assertion. $\square$

The conclusion also controls polynomial and rational approximation of matrix functions. Let $A \in \mathbb{M}_n(\mathbb{C})$, let f be holomorphic on a neighborhood of $W(A)$, and let s be either a polynomial or a rational function pole-free on $W(A)$. Then $f - s$ is holomorphic on a neighborhood of $W(A)$, and Theorem 1.1 gives

$$\|f(A) - s(A)\| \leq 2 \max_{z \in W(A)} |f(z) - s(z)|. \quad (46)$$

This is the sharp universal form of the numerical-range estimate that underlies applications to GMRES and rational Krylov methods; see [7,25,26] for the role of spectral sets in those settings.

7. Discussion and further directions

Theorem 1.1 proves the scalar-valued Crouzeix conjecture. The most immediate strengthening is the uniform matrix-polynomial question: for every $m \geq 1$ and every matrix polynomial $\mathcal{F} = [f_{ij}] \in \mathbb{M}_m(\mathbb{C}[z])$, does one have

$$\|\mathcal{F}(A)\| \leq 2 \max_{z \in W(A)} \|\mathcal{F}(z)\|, \quad \mathcal{F}(A) = [f_{ij}(A)]. \quad (47)$$

Here $\|\mathcal{F}(A)\|$ is the operator norm on $(\mathbb{C}^n)^m$, whereas $\|\mathcal{F}(z)\|$ is the operator norm on $\mathbb{C}^m$. The case $m = 1$ is precisely the scalar theorem proved above; uniformity in m is an additional operator-algebraic requirement. The foundational disk similarity estimate is due to Okubo and Ando [27]; its modern complete-spectral-set formulation and relation to numerical radius are discussed in [28]. Uniform matrix-level spectral-set bounds are studied in [29]; the constant $1 + \sqrt{2}$ at every matrix level follows from [4], while sharp estimates for ρ -contractions with $1 \leq \rho \leq 2$ are developed in [30]. The extension is known with constant 2 for weighted shift matrices [18]. For block-valued inputs, the coefficients in the completion may occupy a larger operator algebra, so the cancellation in (18) would require a genuinely operator-algebraic formulation.

The positive-real completion principle is independent of numerical ranges: it starts with a Carathéodory function whose resolvent defect lies in the adjoint algebra of an auxiliary simple-spectrum matrix sharing an eigenbasis with the matrix in the resolvent. A direct infinite-dimensional extension of Theorem 3.1, based for example on spectral measures or Riesz bases, could reveal equality cases or uniform matrix-level functional-calculus structure. The main challenge is a stable functional-calculus formulation of the auxiliary-basis correction, valid beyond finite diagonal spectral models. Related questions for unbounded numerical ranges are studied in [31], and related abstract spectral-constant problems in [32,33].

Double-layer methods beyond numerical ranges were developed in [7], while configuration-constant refinements were introduced in [23]; see also [22] for a comparison of double-layer approaches. Annular double-layer kernels are studied in [34], while [35] develops a complementary annular

realization and norm theory; elliptic domains admit structured dilation models [36]. Whenever a companion transform lands in a controlled adjoint algebra, the kernel-sampling argument may replace a triangle-inequality estimate and thereby offer a route to sharp constants for other geometric spectral sets.

With the notation of the proof of Theorem 3.1, if $\|\tilde{T}\| = 2$ and $e \in \mathbb{C}^n$ satisfies $\|e\|_2 = 1$ and $\|\tilde{T}e\|_2 = 2$, then (27) and the first series in (23) imply

$$\hat{P}e = 2e, \quad \tilde{T}^2e = 0, \quad \hat{P}\tilde{T}e = \tilde{T}e.$$

Analyzing these vector saturation conditions may classify extremal pairs (A, p) and quantify stability near the nilpotent example (4). Such a classification must allow the nonuniqueness phenomena found in [12].

Finally, O’Loughlin and Rani extend the Crouzeix–Palencia framework to scaled q -numerical ranges, $0 < |q| \leq 1$, and propose associated spectral-set problems [37]. These provide a natural testing ground for the conjectured sharp q -dependent refinement of the constant-2 bound.

AI-assistance disclosure

OpenAI ChatGPT contributed the idea of sampling the matrix Herglotz kernel at half the conjugate eigenvalues and adding the origin sample that cancels the adjoint-algebra correction’s contribution; this appears in the proof of Theorem 3.1, especially (16)–(18). It also assisted with exposition, bibliography, and typesetting. The human author assumes full responsibility for the correctness of all mathematical statements and for the integrity and accuracy of all citations.

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